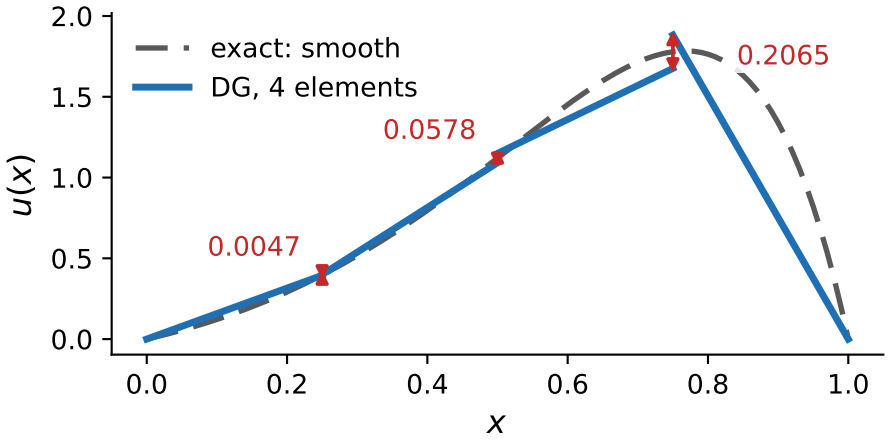
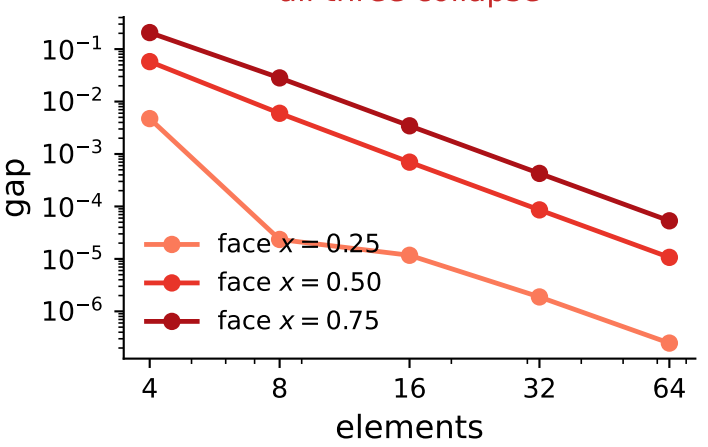
The question is never whether the field jumps. It is whether the TRUE answer jumps there.

A. DG for $-u'' = f$ (second order)

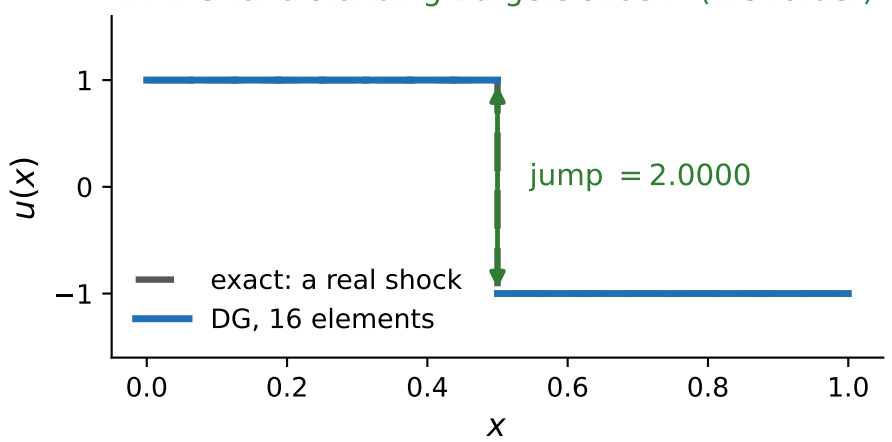


all three collapse

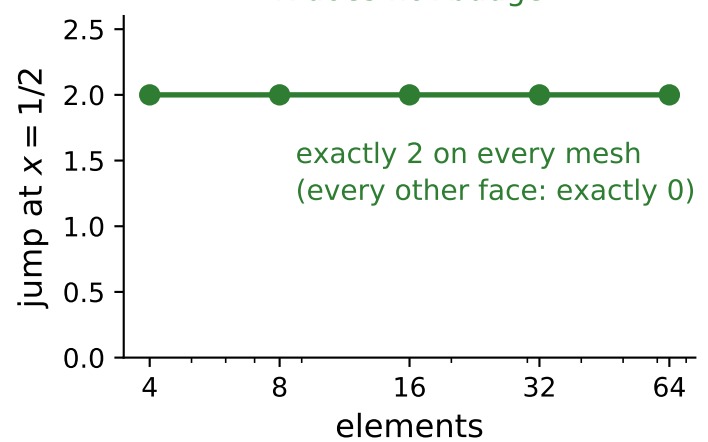


the exact answer has no jump anywhere, so each gap is a deviation from it
⇒ the gap is ERROR, and refinement destroys it

B. DG for a standing Burgers shock (first order)

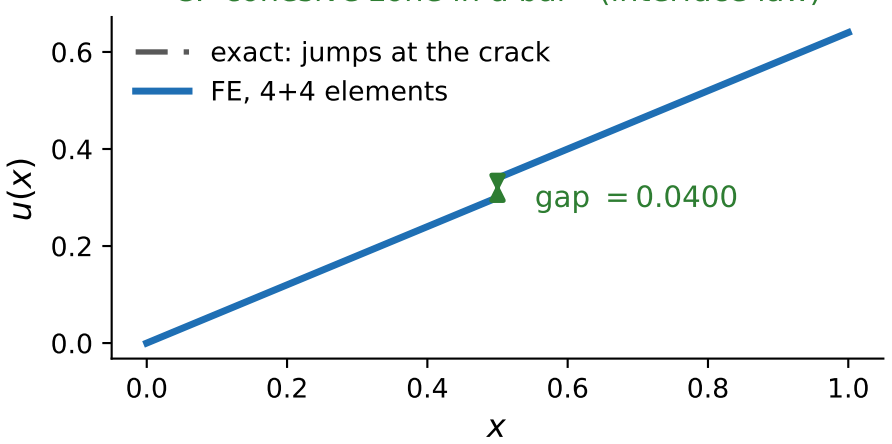


it does not budge

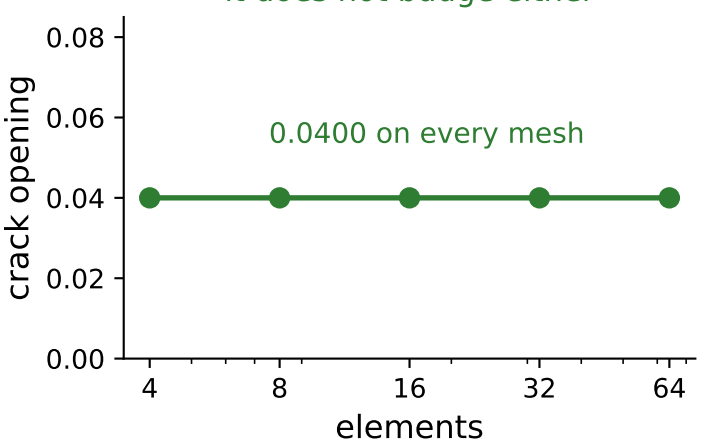


the exact answer really does have this shock, and the scheme reproduces it
⇒ the jump is PHYSICS, and refinement cannot touch it

C. cohesive zone in a bar (interface law)



it does not budge either



the jump is a modelled unknown, with its own law $T(\delta)$ and its own G_c
⇒ the gap is PHYSICS --- it is the crack opening